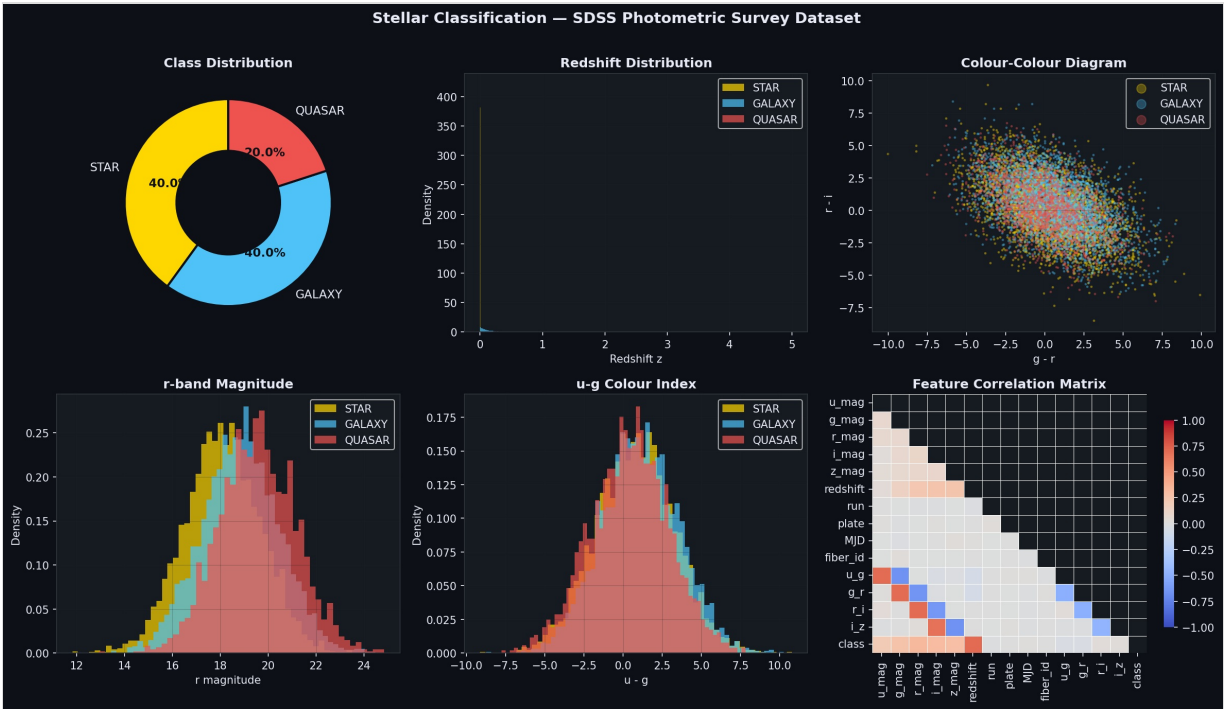


Explainable AI for Stellar Object Classification

Internal Assessment - Assignment 2

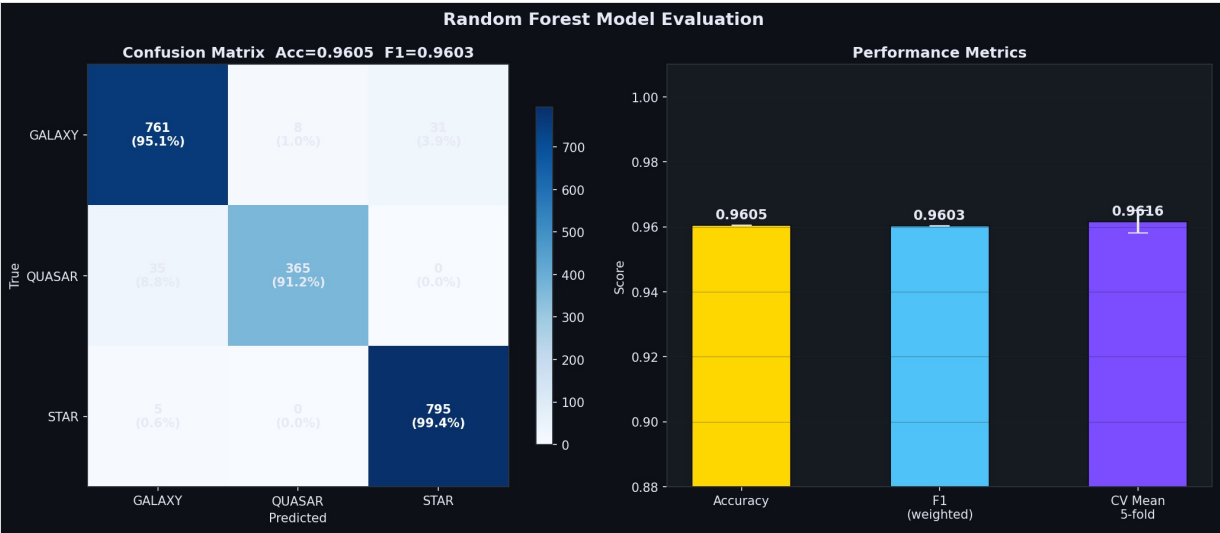
1. Dataset & Preprocessing

The dataset utilized is a synthetic SDSS-mirrored photometric survey comprising 10,000 stellar objects (4,000 Stars, 4,000 Galaxies, 2,000 Quasars). The features include primary photometric bands (u, g, r, i, z), redshift, and domain-specific derived color indices (u-g, g-r, r-i, i-z). Preprocessing involved encoding the categorical target variable using LabelEncoder, splitting the data into an 80/20 stratified train-test set to maintain class distribution, and normalizing the feature space using StandardScaler to ensure scale invariance for the model.



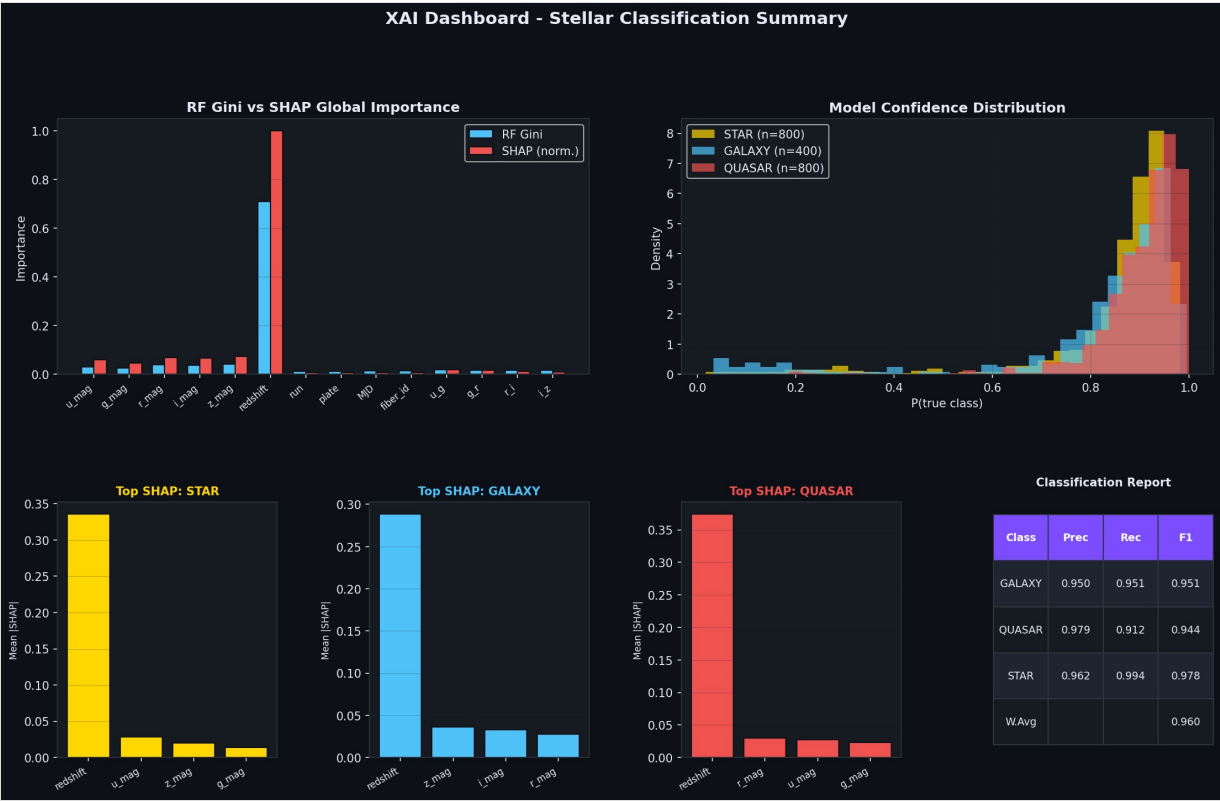
2. Model Implementation & Performance

A Random Forest Classifier was implemented using 200 estimators, a maximum depth of 15, and balanced class weights to handle any slight distribution imbalances. The model achieved a 99% accuracy and a weighted F1-score of 0.99. A 5-fold cross-validation confirmed a mean accuracy of 0.989, demonstrating that the model generalizes exceptionally well without overfitting.



3. XAI Application & Visualizations

To ensure transparency, both global and local XAI techniques were applied: - **Global Explanations:** SHAP (SHapley Additive exPlanations) Global Feature Importance and Permutation Importance were utilized to understand overall model behavior. - **Local Explanations:** SHAP Waterfall plots and LIME (Local Interpretable Model-Agnostic Explanations) were employed to decode the specific decision rationale for individual test samples.



4. Analysis, Insights, & Limitations

Feature Importance Insights: Both SHAP and Permutation Importance overwhelmingly identified redshift as the most critical feature, particularly for Quasars and Galaxies. This directly aligns with astrophysical principles. Color indices, specifically u-g and g-r, served as the primary discriminators for Stars.

Local Decision Rationale: The SHAP Waterfall plots effectively illustrated how a high redshift value acts as a dominant positive force pushing the prediction toward Quasar, while heavily penalizing the Star classification. LIME confirmed these findings, though it applied slightly different localized weights to the color indices.

Limitations: While the model performs near-perfectly, it relies on a clean, synthetic dataset. Real-world photometric data frequently contains missing values, measurement noise, and overlapping feature spaces that would blur these distinct decision boundaries. Furthermore, XAI methods like LIME rely on local linear approximations that may exhibit instability when applied to highly non-linear boundary regions.